Lecture 2

Asking Questions

Announcements

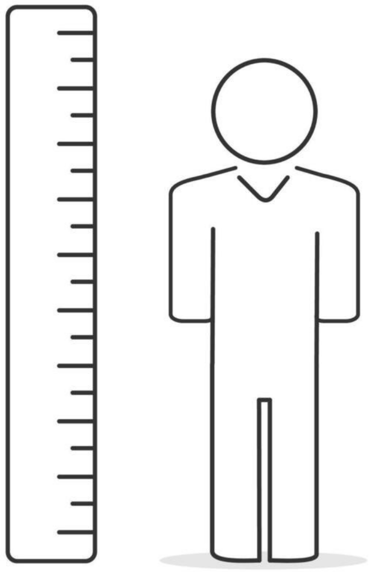
- Lab 0 on Wednesday
  - Getting our computers set up
  - Bring your computer to lab!

Last Time

- Introduced language and ideas for talking about datasets
- Primary focus: tabular data
    - Represent as rows and columns
- Idea: data provenance
    - How/when/where/why/who of our dataset
- Idea: inference
    - Infer things about the world from our dataset
    - an inference is valid if what we see in the dataset is representative of the world

Closing Thoughts

- Imagine you’re given a dataset containing the heights of some Allegheny students
- You calculate the mean height
- Is this mean a good estimate of the average height of Allegheny students?
- When might it be? When might it not?



Today

- Survey summary statistics
- Asking questions about a dataset
  - Description
  - Comparison
  - Association

Summary Statistics

- Common ways of summarizing a variable:
  - Total (or sum)
  - Mean (or average)
  - Maximum/Minimum (or range)
  - Median (or middle)
  - Other percentiles
    - Example: 90th percentile if we're looking at exam scores
  - Variance
    - How far a set of numbers is spread out from its average

Descriptive Questions

- **Ask**: "what is happening with this variable?"
- **Goal**: summarize individual variables
- **Examples**:
  - How many people visited our course website last week?
  - What is the 90^{th} percentile SAT score?
  - How much do students study on average?

Comparative Questions

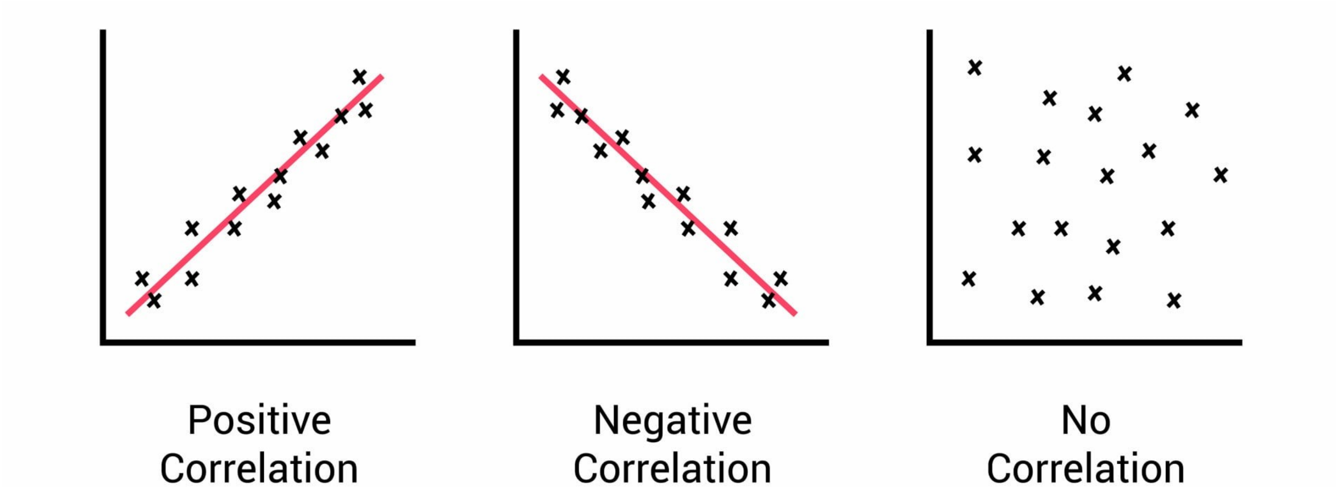
- **Ask:** "how do these groups differ?"
- **Goal:** summarize differences between different groups in dataset
- **Examples**:
  - Did more Firefox or Safari users visit the website last week?
  - Did PA or OH students have a higher 90^{th} percentile SAT score?
  - Do students who go to office hours have a higher or lower GPA?

Associative Questions

- **Ask**: "how are these variables connected?"
- **Goal**: summarize patterns in the dataset
- **Examples**:
    - How does number of website visits vary by iOS version?
    - What's the relationship between distance from test location and SAT score?
    - Do students who study more have higher GPAs?

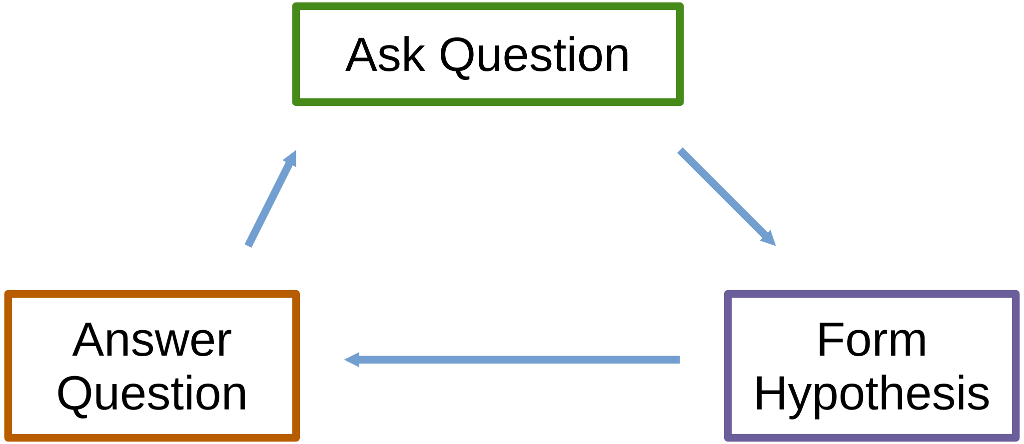
Associative Questions

- Tell us how variables *correlate*
  - Positive correlation means as one increases the other increases
  - Negative correlation means as one increases the other decreases
- Measuring and examining correlations is a major part of data analysis



Questions and Hypotheses

- Feedback loop between answering questions and forming hypotheses



Exercise

Imagine you have a dataset containing daily swipe data for the dining halls

- Variables:
  - date, dining hall, # swipes, featured item, semester week, coffee consumed

Formulate a descriptive, a comparative, and an associative question

- Offer a hypothesis about what you expect to find

Next Time

- Read CIT Ch. 2
- Correlation vs. Causation